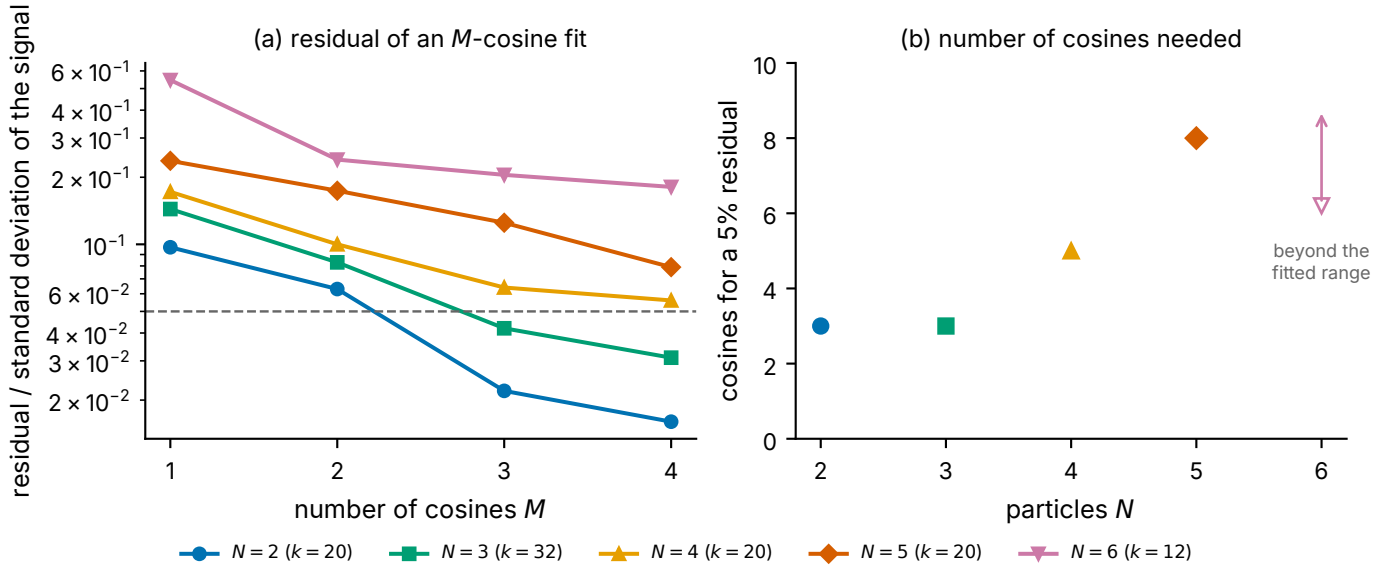
(a) residual of an M-cosine fit
residual / standard deviation of the signal
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
1
2
3
4
number of cosines M
(b) number of cosines needed
cosines for a 5% residual
10
8
6
4
2
0
2
3
4
5
6
particles N
beyond the
fitted range
$N = 2$ ($k = 20$)
$N = 3$ ($k = 32$)
$N = 4$ ($k = 20$)
$N = 5$ ($k = 20$)
$N = 6$ ($k = 12$)